

A054 · Integrative Radiogenomic Analysis Identifies Imaging-Linked Molecular Subtypes and Biomarkers in Pancreatic Ductal Adenocarcinoma

Presenter: Zhi Qu — Department of Radiation Oncology, University of Rochester Medical Center, Rochester, NY

Authors: Paul M Grandgenett, Michael A Hollingsworth

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Pancreatic ductal adenocarcinoma (PDAC) exhibits substantial molecular heterogeneity associated with differences in tumor biology and clinical outcomes. Integrating multi-omics and imaging datasets holds promise for refining subtype identification and therapeutic stratification. Non-Negative Matrix Factorization (NMF) of the TCGA pancreatic adenocarcinoma cohort (TCGA-PAAD) identified six transcriptomic clusters, which were characterized against established Bailey molecular subtypes. Clusters 2, 4, and 6 were enriched for ADEX, Immunogenic, and Squamous subtypes, respectively. A Random Forest classifier with iterative SHAP-guided feature selection identified a 35-marker-gene panel. The classifier was then applied to an independent rapid-autopsy cohort (RAP) from the University of Nebraska Medical Center (UNMC) with matched transcriptomic and contrast-enhanced CT imaging. The RF model assigned RAP patients to molecular subtypes with distinct survival patterns. Notably, patients assigned to Cluster 2 (ADEX-like) experienced longer survival, whereas patients in Cluster 6 (Squamous) had the shortest survival. Exploratory radiogenomic analysis revealed significant associations between the 35 marker genes and 944 CT radiomic features. Specifically, the squamous-like marker genes *EPHA2* and *NTF4* showed a negative correlation with texture complexity metrics *wavelet-HLH_glszm_SizeZoneNonUniformityNormalized* and *wavelet-HHL_ngtdm_Contrast*, respectively. Ultimately, integrating molecular and radiomic markers suggests that non-invasive CT imaging can act as a surrogate for underlying tumor biology. High expression of squamous markers (*EPHA2*/*NTF4*) coupled with a low size-zone uniformity or low neighboring contrast on CT reflects a shift toward coarse, poorly perfused, or necrotic tissue. This distinct radiogenomic profile serves as a non-invasive indicator of a highly aggressive squamous-like phenotype and poor patient prognosis.

Integrative Radiogenomic Analysis Identifies Imaging-Linked Molecular Subtypes and Biomarkers in Pancreatic Ductal Adenocarcinoma

Zhi Qu^{1,2}, Paul M Grandgenett³, Mike Baine⁴, Qian Du⁵, Michael A Hollingsworth⁶, Chi Zhang^{5,*},
Dandan Zheng^{1,*}

1. Department of Radiation Oncology, University of Rochester Medical Center, Rochester, NY
2. Department of Biomedical Informatics, Harvard Medical School, Boston, MA
3. Rapid Autopsy Program, University of Nebraska Medical Center, Omaha, NE
4. Department of Radiation Oncology, University of Nebraska Medical Center, Omaha, NE
5. School of Biological Sciences, University of Nebraska, Lincoln, NE
6. The Eppley Institute for Research in Cancer and Allied Diseases, University of Nebraska Medical Center, Omaha, NE

Background: Pancreatic ductal adenocarcinoma (PDAC) is characterized by high molecular and structural heterogeneity, which underlies significant variation in tumor biology and patient outcomes. Integrating multi-omics and imaging datasets holds promise for refining subtype identification and therapeutic stratification.

Methods: Using Non-Negative Matrix Factorization (NMF) on the TCGA pancreatic adenocarcinoma cohort (TCGA-PAAD), we identified six transcriptomic clusters. These were characterized against established Bailey molecular subtypes: Cluster 2 demonstrated enrichment for the Aberrantly Differentiated Endocrine Exocrine (ADEX) subtype, Cluster 4 for the Immunogenic subtype, and Cluster 6 for the highly aggressive Squamous subtype. To construct a robust molecular classifier, we trained a Random Forest (RF) model using an initial set of 4,075 non-redundant genes. Iterative SHAP-guided recursive feature elimination (RFE) successfully reduced this signature to a highly discriminative 35-marker-gene panel, yielding multi-class cross-validation AUC values above 0.949. The final classifier was validated on an independent rapid-autopsy cohort (RAP) from the University of Nebraska Medical Center (UNMC) with matched transcriptomic and contrast-enhanced CT imaging.

Results: In the RAP cohort, the RF model successfully assigned patients to molecular subtypes, demonstrating differences in overall survival. Notably, patients assigned to Cluster 2 (ADEX-like) experienced longer survival, whereas patients in Cluster 6 (Squamous) had the shortest survival. Exploratory radiogenomic analysis revealed significant associations between the 35 marker genes and 944 CT radiomic features. Specifically, the squamous-like marker genes *EPHA2* and *NTF4* showed a negative correlation with texture complexity metrics wavelet-HLH_glszm_SizeZoneNonUniformityNormalized and wavelet-HHL_ngtdm_Contrast, respectively.

Conclusions: Ultimately, integrating molecular and radiomic markers suggests that non-invasive CT imaging can act as a surrogate for underlying tumor biology. High expression of squamous markers (*EPHA2/NTF4*) coupled with a low size-zone uniformity or low neighboring contrast on CT reflects a shift toward coarse, poorly perfused, or necrotic tissue. This distinct radiogenomic profile serves as a non-invasive, high-confidence indicator of a highly aggressive squamous-like phenotype and poor patient prognosis